Solar Wind Mass-Energy Flux Partitioning Budget in the UQFF Heliosphere

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2026-04-15

PAPER_1078: Solar Wind Mass-Energy Flux Partitioning Budget in the UQFF Heliosphere

Abstract

We derive a complete mass-energy flux partitioning budget for stellar wind propagation through the UQFF heliosphere. The total solar wind flux $\Phi_{\text{total}} = 4\pi r_{\text{helio}}^2 \rho_{\text{sw}} v_{\text{sw}}$ is decomposed into four channels: (a) planetary geometric interception f_{planet}, (b) Ug2 outer-field-shell transmutation into hydrogen complexes f_{Ug2}, (c) planetary core absorption f_{core}, and (d) heliospheric escape f_{escape}. Numerical evaluation for the 8-planet solar system yields $f_{\text{Ug2}} \approx 0.85$, confirming the dominant role of Ug2 charge-reactivity in processing solar wind material into magnetically bound hydrogen complexes on the outer heliospheric shell.

§1 Total Wind Flux

$$\Phi_{\text{total}} = 4\pi r_{\text{helio}}^2 \cdot \rho_{\text{sw}} \cdot v_{\text{sw}}$$

At 1 AU: $\rho_{\text{sw}} \approx 8 \times 10^{-21}$ kg/m^3, $v_{\text{sw}} \approx 5 \times 10^5$ m/s.

§2 Density Scaling

Solar wind density falls as inverse-square:

$$\rho_{\text{sw}}(d) = \rho_{\text{sw,1AU}} \cdot \left(\frac{\text{AU}}{d}\right)^2$$

§3 Per-Planet Interception

Each planet intercepts flux through its magnetospheric cross-section:

$$\Phi_{\text{planet},i} = \pi R_{\text{mag},i}^2 \cdot \rho_{\text{sw}}(d_i) \cdot v_{\text{sw}}$$

$$f_{\text{planet}} = \frac{\sum_i \Phi_{\text{planet},i}}{\Phi_{\text{total}}}$$

Planet	R_{mag} (m)	d (AU)	$\Phi_i/\Phi_{\text{total}}$
Mercury	1.5×10^6	0.387	$\sim 10^{-14}$
Venus	6.1×10^6	0.723	$\sim 10^{-13}$
Earth	6.4×10^7	1.0	$\sim 10^{-11}$
Mars	3.4×10^6	1.524	$\sim 10^{-14}$
Jupiter	7.1×10^{10}	5.203	$\sim 10^{-5}$
Saturn	2.0×10^{10}	9.537	$\sim 10^{-6}$
Uranus	1.8×10^9	19.19	$\sim 10^{-8}$
Neptune	2.4×10^9	30.07	$\sim 10^{-8}$

§4 Four-Channel Budget

$$f_{\text{core}} = f_{\text{planet}} \cdot \eta_{\text{penetration}} \cdot (1 - \eta_{\text{liquid}})$$

$$f_{\text{Ug2}} = (1 - f_{\text{planet}}) \cdot \eta_{\text{transmutation}}$$

$$f_{\text{escape}} = 1 - f_{\text{planet}} - f_{\text{Ug2}}$$

With $\eta_{\text{transmutation}} = 0.85$, $\eta_{\text{penetration}} = 0.15$, $\eta_{\text{liquid}} = 0.6$:

Channel	Fraction	Physical Process
f_{Ug2}	≈ 0.85	Transmutation $\rightarrow$ H complexes on outer shell
f_{escape}	≈ 0.15	Escape past heliosphere boundary
f_{planet}	$\sim 10^{-5}$	Geometric interception by magnetospheres
f_{core}	$\sim 10^{-6}$	Absorbed into planetary cores (maintains Um)

§5 Simulation Sets

- **High wind:** $\rho_{\text{sw}} = 1.6 \times 10^{-20}$, $v_{\text{sw}} = 8 \times 10^5$ m/s
- **Low wind:** $\rho_{\text{sw}} = 4 \times 10^{-21}$, $v_{\text{sw}} = 3 \times 10^5$ m/s
- **Young star:** $\rho_{\text{sw}} = 5 \times 10^{-20}$, $v_{\text{sw}} = 1 \times 10^6$ m/s

§6 UQFF Connection

The dominant Ug2 channel ($f_{\text{Ug2}} \approx 0.85$) validates the Star-Magic prediction that solar winds not absorbed by planets are transmuted by the Ug2 outer field shell into hydrogen complexes magnetically stuck to the heliosphere boundary, providing the raw material for Ug2 charge-reactivity processes described in PAPER_412.

References

- PAPER_412: HeliosphereHydrogenComplexSCmStellarAgeCalculator
- PAPER_413: Ug3CCWCWDifferentialRotationSCmPlanetaryCoreCalculator
- Star-Magic.txt: §3 Solar Wind and Heliospheric Flux Balance

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger.* Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102
2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series.* Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic